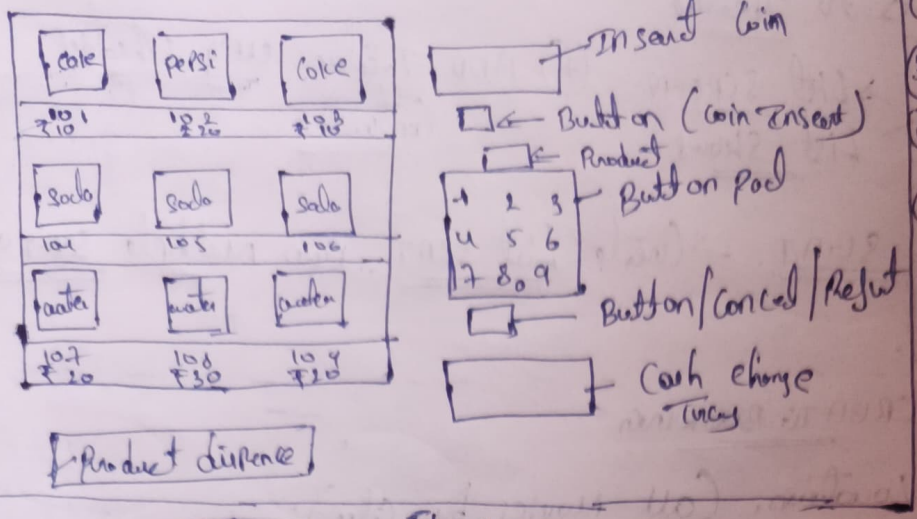
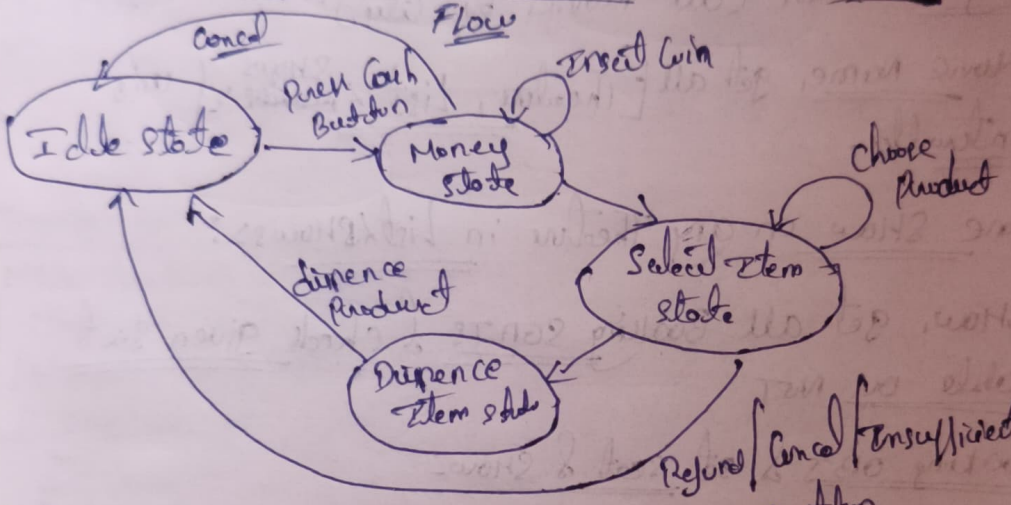
(8) VENDING MACHINE = STATE PATTERN



① Insert Coin button
② Insert Coin
③ Select Product
④ Enter Code
⑤ Press Ok
⑥ Cancel Button
⑦ get Refund amount



OBSERVATION

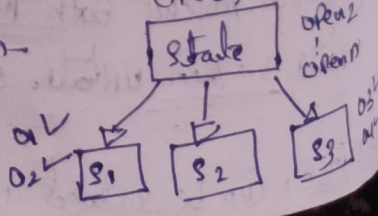
→ There are different set of states & on each state there is different operations.

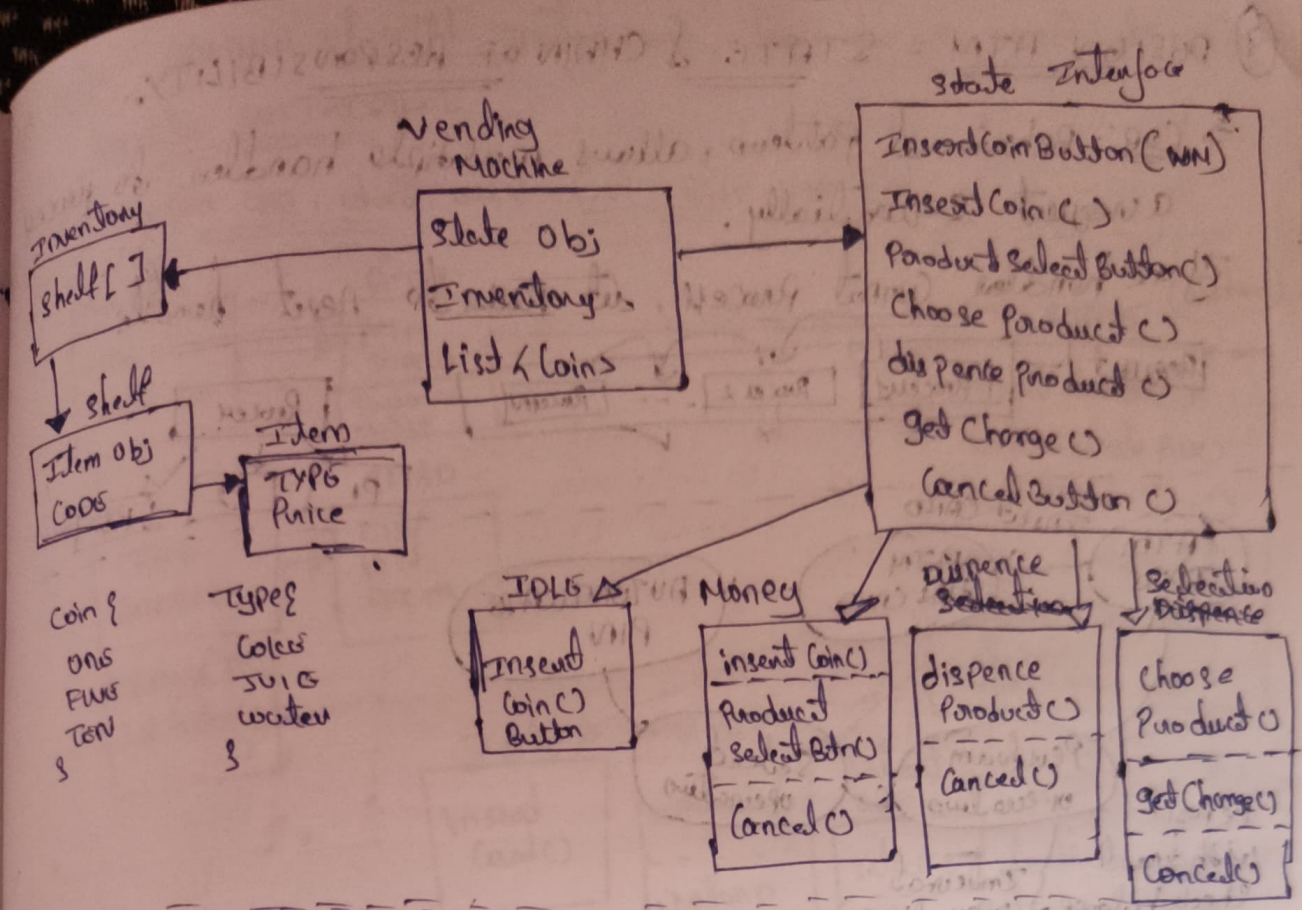
① Idle → Press Insert Coin Button
② Money → Insert Coin, select Product Button, Cancel/Refund.
③ select Product → choose Product, Cancel/Refund, Return Change
④ Dispence → Product dispence

→ STATE Pattern: Allows object to change its state, behaviour when internal state changes

→ Instead of if-else, we use state pattern

* Depending on state, some operation works & some throw exceptions





FLOW

① Vending Machine has Vending state → Idle initially

② It has also a Inventory: Inventory has List<Shelf>

③ Shelf → has Item & Code. Item is having Price & Type.

④ Fill up the Inventory. Fill Away with "Item & Code"

⑤ → State = Idle → Insert Coin Btn (VM) → State = Money State (S)

→ State = Money → Insert Coin() → Add coins to Coin List

Click Product Select Btn() → state = Selection

→ State = Selection → Chose Product (code) → Done dispence

→ State = Dispence → Dispense() → Idle state

⑥ Repeat Process.